



# LAB 04-MANAGING EXCHANGE DATABASE



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### Exercise 1 – Managing an Exchange 2019 Database using EAC

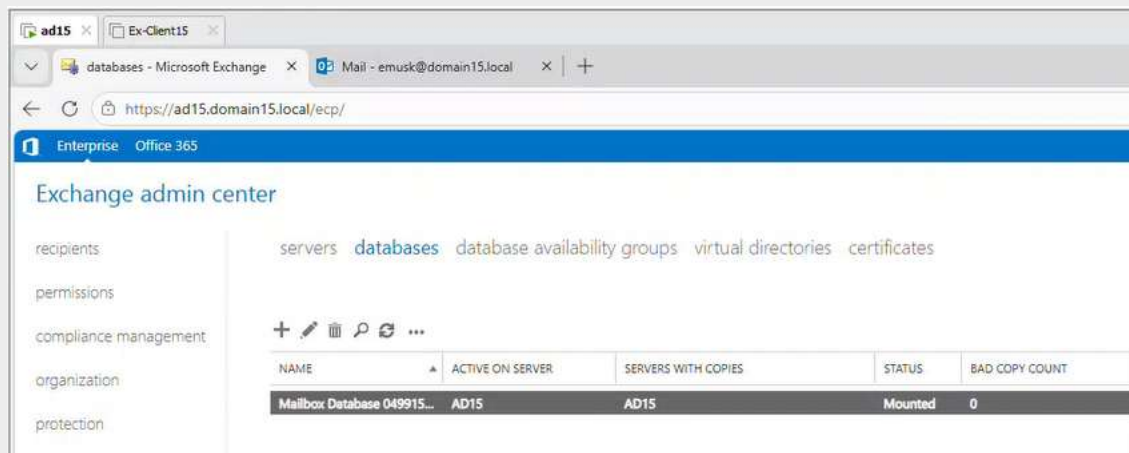
Objective:

- Manage the Exchange 2019 database using **Exchange Administration Center (EAC)**

#### Tasks:

##### Part 1: Database Overview and Status

1. Open the **Exchange Administration Center (EAC)** and navigate to **Servers > Databases**.



2. Select an existing database and review its properties:

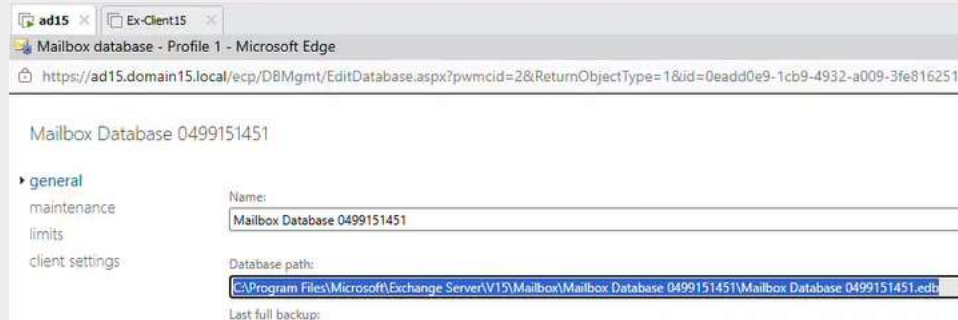
- What is the **status** of the database?

**Mounted.**

- What is the **server name** hosting this database?

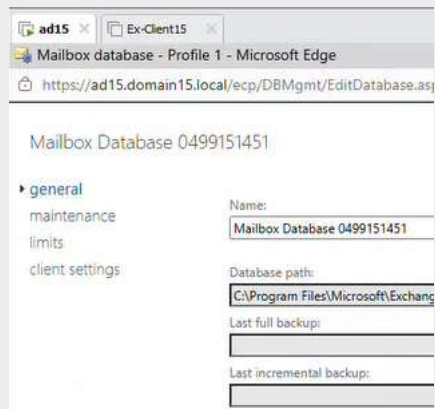
**AD15**

- What is the **database path**?



- What is the date of the **last full** and last **incremental** backup?

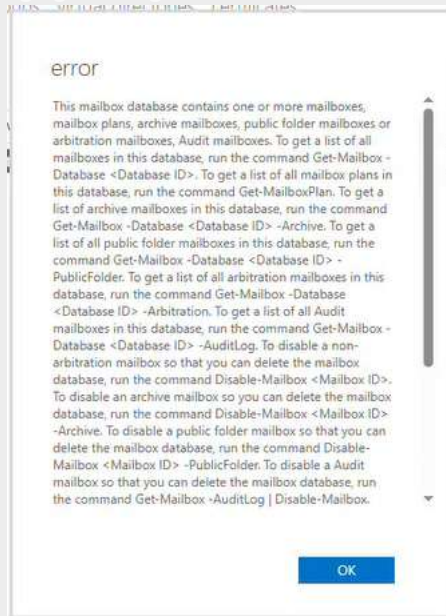
**Shows no date.**



3. Try to **delete** this database from the EAC.

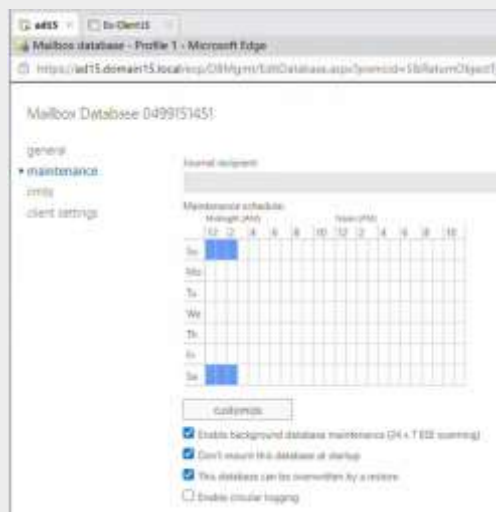
- Are you able to do so? Why or why not?

**No, I am not able to delete the Administrator database because it still contains one or more mailboxes. Exchange does not allow a mailbox database to be removed while user mailboxes, archive mailboxes, arbitration mailboxes, monitoring mailboxes, or other system mailboxes still exist in the database.**



## Part 2: Maintenance Settings

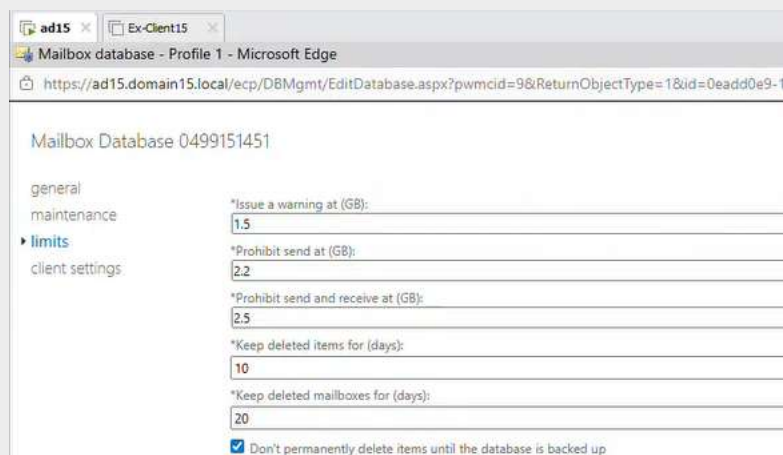
4. Open the **properties** of the same database and configure the following:
  - Modify the **maintenance schedule** to run only on **weekends** between **midnight and 2:00 AM**.
  - Enable the option **“This database can be overwritten by a restore”**.
  - Enable **background maintenance** and **circular logging**.
  - Prevent the database from mounting at **startup**.



### Part 3: Storage Limits

5. In the **Limits** section of the database settings:

- Set the warning threshold to **1.5 GB**.
- Set the prohibit send limit to **2.2 GB**.
- Set the prohibit send and receive limit to **2.5 GB**.
- Retain deleted items for **10 days**.
- Retain deleted mailboxes for **20 days**.
- Enable the option **“Don’t permanently delete items until the database is backed up”**.



Mailbox Database 0499151451

|                 |  |
|-----------------|--|
| general         |  |
| maintenance     |  |
| limits          |  |
| client settings |  |

\*Issue a warning at (GB): 1.5

\*Prohibit send at (GB): 2.2

\*Prohibit send and receive at (GB): 2.5

\*Keep deleted items for (days): 10

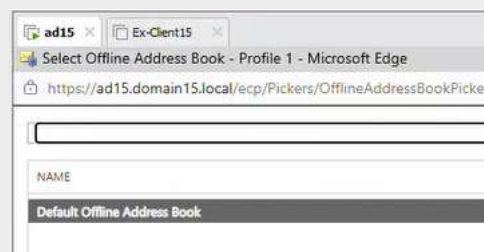
\*Keep deleted mailboxes for (days): 20

☒ Don't permanently delete items until the database is backed up

### Part 4: Client Settings

6. In the **Client Settings** tab:

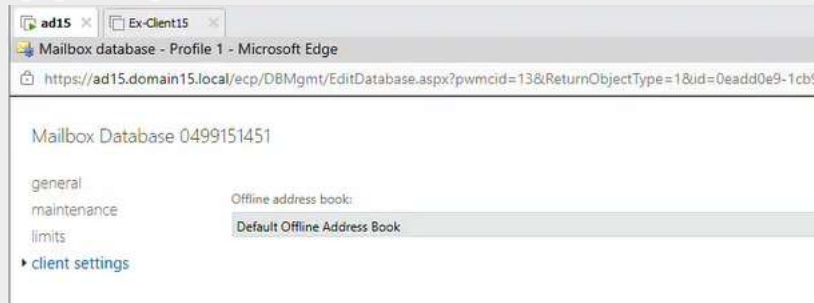
- Change the Offline Address Book (OAB) assigned to the database.
- What options are available for OAB selection?



Select Offline Address Book - Profile 1 - Microsoft Edge

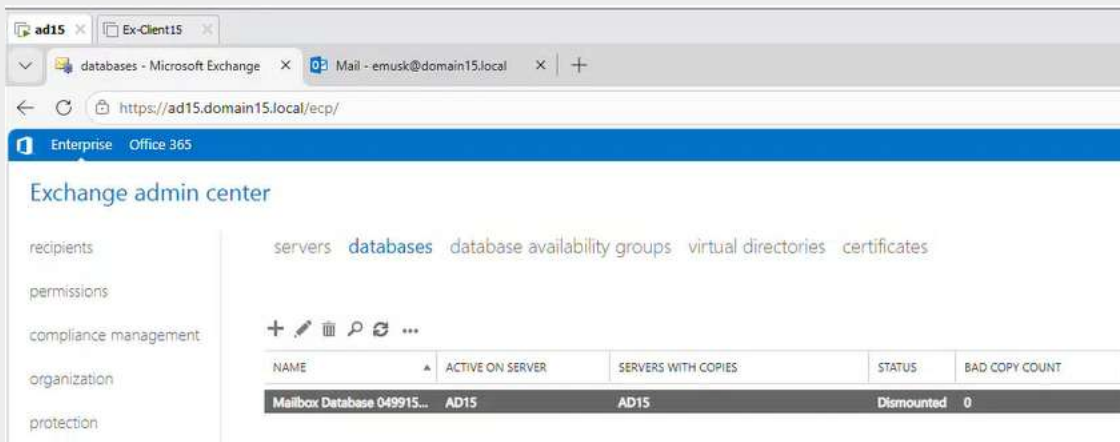
https://ad15.domain15.local/ecp/Pickers/OfflineAddressBookPicke

|                              |
|------------------------------|
| NAME                         |
| Default Offline Address Book |

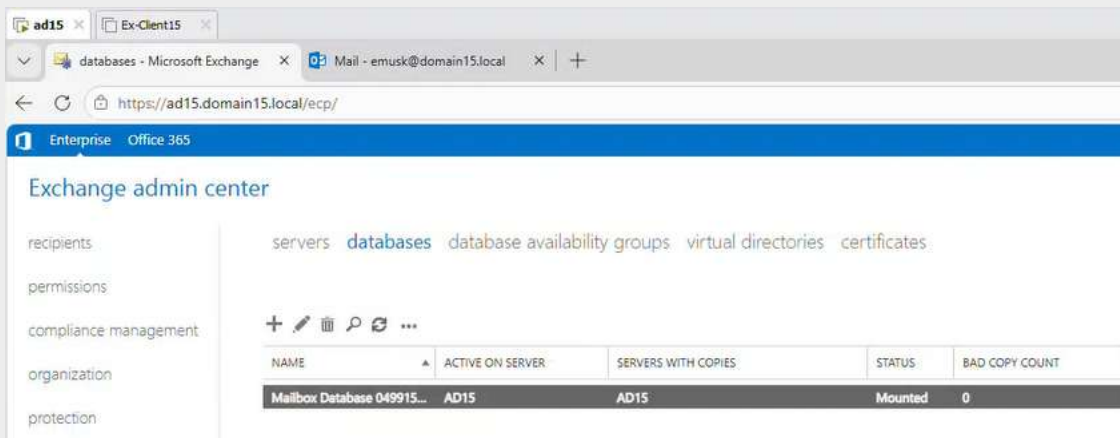


## Part 5: Mounting and Dismounting

7. Select the database and **dismount** it from the EAC.
8. Refresh the status and verify that the database is no longer mounted.



9. Now **mount** the same database again.



**\*\* Close the Exchange Administration Center (EAC) console.**

## Exercise 2 – Manage Exchange 2019 Database using PowerShell (EMS)

**Objective:**

Perform **mailbox database administration tasks** in Exchange using **PowerShell (EMS)**.

**Tasks:****Part 1: Database Information and Status**

1. Display all mailbox databases hosted by your server **adXX**.

```

Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase -server ad15

Name                Server      Recovery      ReplicationType
----                -
Mailbox Database 0499151451  AD15        False         None
  
```

2. Display all properties of this database.

```

Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase "Mailbox Database 0499151451" | Format-List

RunspaceId                : 29f7bdec-2fb0-4c2f-94b4-6491ce513d9f
JournalRecipient           :
MailboxRetention           : 20.00:00:00
OfflineAddressBook         : \Default Offline Address Book
OriginalDatabase           :
PublicFolderDatabase       :
ProhibitSendReceiveQuota   : 2.5 GB (2,684,354,560 bytes)
ProhibitSendQuota          : 2.2 GB (2,362,232,832 bytes)
RecoverableItemsQuota      : 30 GB (32,212,254,720 bytes)
RecoverableItemsWarningQuota : 20 GB (21,474,836,480 bytes)
CalendarLoggingQuota       : 6 GB (6,442,450,944 bytes)
IndexEnabled               : True
IsExcludedFromProvisioning  : False
IsExcludedFromProvisioningReason :
IsExcludedFromProvisioningBy :
IsExcludedFromProvisioningBySchemaVersionMonitoring : False
IsExcludedFromInitialProvisioning : False
IsSuspendedFromProvisioning : False
IsExcludedFromProvisioningBySpaceMonitoring : False
IsExcludedFromProvisioningForDraining : False
IsExcludedFromProvisioningByOperator : False
IsExcludedFromProvisioningDueToLogicalCorruption : False
MailboxLoadBalanceMaximumEdbFileSize :
MailboxLoadBalanceRelativeLoadCapacity :
MailboxLoadBalanceOverloadedThreshold :
MailboxLoadBalanceUnderloadedThreshold :
MailboxLoadBalanceEnabled :
DumpsterStatistics         :
DumpsterServersNotAvailable :
ReplicationType            : None
AdminDisplayVersion        : Version 15.2 (Build 1748.10)
AdministrativeGroup         : Exchange Administrative Group (FYDIBOHF23SPDLT)
AllowFileRestore           : False
BackgroundDatabaseMaintenance : True
  
```

3. Display only the name and GUID of this database.

```

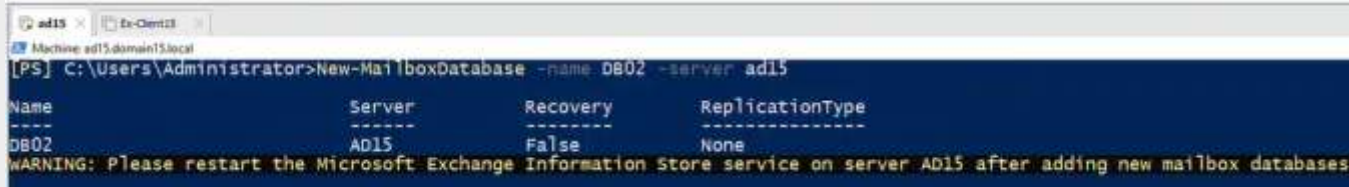
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase "Mailbox Database 0499151451" | fl name, guid

Name : Mailbox Database 0499151451
Guid : 0eadd0e9-1cb9-4932-a009-3fe8162511fc
  
```

**Part 2: Creating and Managing Databases**



4. Create a new mailbox database named **DB02** on the server **adXX** using default settings.

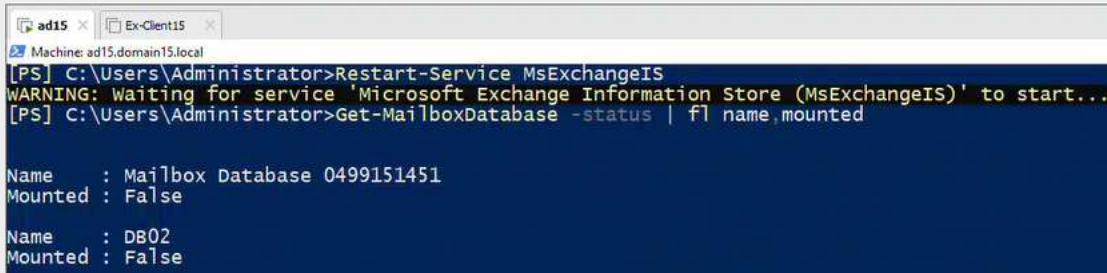


```
[PS] C:\Users\Administrator>New-MailboxDatabase -name DB02 -server ad15

Name          Server      Recovery    ReplicationType
----          -
DB02          AD15        False       None

WARNING: Please restart the Microsoft Exchange Information Store service on server AD15 after adding new mailbox databases
```

5. Restart the Exchange Information Store service to refresh memory allocation.
6. Verify if the mailbox databases are mounted.



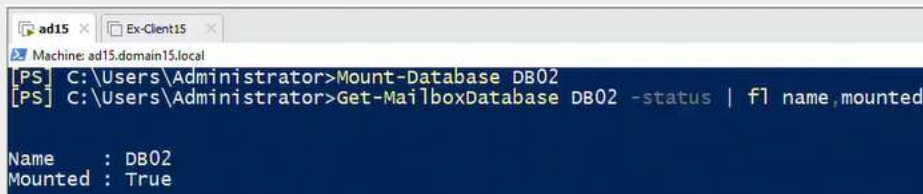
```
[PS] C:\Users\Administrator>Restart-Service MsExchangeIS
WARNING: Waiting for service 'Microsoft Exchange Information Store (MsExchangeIS)' to start...
[PS] C:\Users\Administrator>Get-MailboxDatabase -status | fl name,mounted

Name          : Mailbox Database 0499151451
Mounted       : False

Name          : DB02
Mounted       : False
```

### Part 3: Mounting and Dismounting

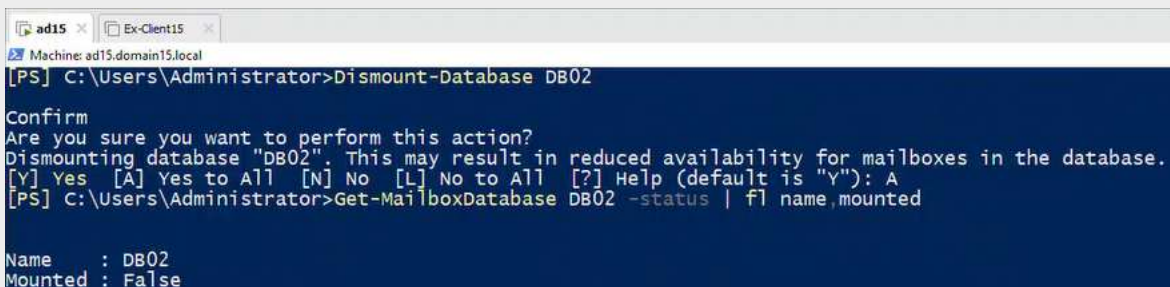
7. Mount the mailbox database **DB02**.
8. Verify its mount status.



```
[PS] C:\Users\Administrator>Mount-Database DB02
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 -status | fl name,mounted

Name          : DB02
Mounted       : True
```

9. Dismount the **DB02** database.
10. Verify its status again.



```
[PS] C:\Users\Administrator>Dismount-Database DB02

Confirm
Are you sure you want to perform this action?
Dismounting database "DB02". This may result in reduced availability for mailboxes in the database.
[Y] Yes [A] Yes to All [N] No [L] No to All [?] Help (default is "Y"): A
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 -status | fl name,mounted

Name          : DB02
Mounted       : False
```

---

### Part 4: File Path Management

11. Create the folder **C:\Databases\DB02**.
12. Move the **DB02** database file to **C:\Databases\DB02** and rename it db002.edb.



13. Move the transaction log files of **DB02** to the same folder.

```

ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>mkdir "c:\Databases\DB02"

Directory: C:\Databases

Mode                LastWriteTime         Length Name
----                -
d-----          6/6/2026 11:57 AM             DB02

[PS] C:\Users\Administrator>Move-DatabasePath DB02 -EdbFilePath "c:\Databases\DB02\DB002.edb"

Confirm
Are you sure you want to perform this action?
Moving database path "DB02":
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [?] Help (default is "Y"): A
[PS] C:\Users\Administrator>Move-DatabasePath DB02 -LogFolderPath "c:\Databases\DB02\"

Confirm
Are you sure you want to perform this action?
Moving database path "DB02":
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [?] Help (default is "Y"): A
[PS] C:\Users\Administrator>
  
```

14. Display the updated database and log file paths.

```

ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 | Format-List EdbFilePath, LogFolderPath

EdbFilePath      : c:\Databases\DB02\DB002.edb
LogFolderPath     : c:\Databases\DB02
  
```

## Part 5: Quota Configuration

15. Mount the **DB02** database again.

```

ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Mount-Database DB02
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 -status | fl name,mounted

Name      : DB02
Mounted   : True
  
```

16. Set a warning quota of **800 MB** on the **DB02** database.
17. Set a prohibit send quota of **2 GB**.
18. Remove the prohibit send/receive quota limits.

```

ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 | Set-MailboxDatabase -IssueWarningQuota 800MB
[PS] C:\Users\Administrator>Set-MailboxDatabase DB02 -ProhibitSendQuota 2GB
[PS] C:\Users\Administrator>Set-MailboxDatabase DB02 -ProhibitSendReceiveQuota unlimited
[PS] C:\Users\Administrator>
  
```

19. Verify the new quota settings.

## Lab 04 - Managing Exchange 2019 Database

```
Machine: ad15.demon15.local
[PS] C:\Users\Administrator>Get-MailboxDatabase DB02 | Format-List IssueWarningQuota, ProhibitSendQuota, ProhibitSendReceiveQuota, DeletedItemRetention, MailboxRetention

IssueWarningQuota      : 800 MB (838,860,800 bytes)
ProhibitSendQuota       : 2 GB (2,147,483,648 bytes)
ProhibitSendReceiveQuota : Unlimited
DeletedItemRetention    : 14.00:00:00
MailboxRetention        : 30.00:00:00
```

\* Delete the Database **DB02**.

```
Machine: ad15.demon15.local
[PS] C:\Users\Administrator>Remove-MailboxDatabase DB02

Confirm
Are you sure you want to perform this action?
Removing mailbox database "DB02".
[Y] Yes [A] Yes to All [N] No [L] No to All [?] Help (default is "Y"): a
WARNING: The specified database has been removed. You must remove the database file located in c:\Databases\DB02\DB002.edb from your computer manually if it exists. Specified database: DB02
[PS] C:\Users\Administrator>get-mailboxdatabase

Name                               Server      Recovery      ReplicationType
----                               -
Mailbox Database 0499151451        AD15        False         None
```